

SWAPNIL KOLE

Senior Software Engineer | Blockchain & Full-Stack Developer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience building full-stack, decentralized, and AI-powered applications across the MERN stack, Solidity smart contracts, and cloud infrastructure (AWS, Azure, GCP). Specializes in architecting NFT and Web3 platforms end-to-end — from smart contract development, testing, and auditing to scalable backend systems and React front ends. Proven record of leading engineering teams, conducting code reviews, and independently delivering full-stack and blockchain solutions end-to-end for enterprise clients, startups, and government bodies.

CORE SKILLS

Languages: JavaScript (ES6), TypeScript, Solidity, Python, PHP, Go, C++, C, SQL

Frontend: React, Next.js, HTML5, CSS3, SCSS, Bootstrap, Material UI, ReactStrap, jQuery

Backend: Node.js, Express.js, PHP, REST APIs, Axios

Blockchain & Web3: Solidity, Hardhat, Truffle, Web3.js, IPFS, Hyperledger, T-REX (ERC-3643), Staking Contracts, Polygon, Smart Contract Auditing, dApp Development

Databases: MongoDB, MySQL, SQL

Cloud: AWS (CDK), Microsoft Azure, Google Cloud Platform (GCP)

DevOps & IaC: Terraform, AWS CDK, Infrastructure as Code, CI/CD

AI & Integrations: AI/LLM Integration, Recommendation Systems, Amazon API, Telegram Mini Apps

Tools & Practices: GitHub, Code Review, System Design, Agile

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Vistateq Virtuosity

Oct 2025 – Present

India • Client: BridgetowerCapital

Technologies: AWS CDK, Terraform, Solidity (T-REX / ERC-3643), React

- Architected and built the complete backend infrastructure for an NFT platform using AWS CDK.
- Provisioned and automated cloud infrastructure as code with Terraform.
- Developed T-REX (ERC-3643) compliant smart contracts, including deployment scripts and automated test suites.
- Integrated the React front end with backend services and UI components, and designed end-to-end system flows.

Lead Engineer — SuperWorld

May 2022 – Sep 2025

California, USA

Technologies: MERN (MongoDB, Express, React, Node.js), Solidity, Web3.js, Polygon

- Built a virtual real estate map and NFT marketplace web application on the MERN stack.
- Developed, deployed, tested, and audited Solidity smart contracts on the Polygon network.
- Integrated smart contracts with the MERN application using Web3.js.
- Implemented payment gateways and third-party service integrations.
- Led a team of 5 junior engineers and conducted code reviews of their work.

FREELANCE & CONTRACT EXPERIENCE

Full-Stack & AI Developer — GetFi

5 months

Telegram-based e-commerce, rewards & gaming product

Technologies: Telegram Mini App, AI Recommendation System, Amazon API, Full-Stack

- Built a Telegram mini-app product combining gamified rewards, user engagement, and in-app shopping.
- Designed and built an AI-powered, conversational product recommendation engine — a “ChatGPT for shopping” that surfaces relevant products in real time, integrated with the Amazon API.
- Delivered the project end-to-end across the front end, back end, and AI recommendation system.

Full-Stack Blockchain Developer — BlockX

8 months

Singapore

Technologies: Solidity, React, Smart Contracts, BlockX Chain

- Designed and developed staking smart contracts, then tested and deployed them on the BlockX chain.
- Built a complete decentralized application (dApp) integrating React with on-chain blockchain logic.

Technical Blockchain Assistant — IIIT Bhubaneswar

20 months

Government of Odisha, Dept. of Science and Technology • Bhubaneswar, India

Technologies: Blockchain, Solidity, Smart Contracts, dApps

- Designed and built a Land Record Management application for the Government of Odisha.
- Developed all core algorithms underpinning the system, and tested and deployed the smart contracts.
- Delivered a full decentralized application (dApp) on blockchain and published a peer-reviewed conference paper.

Web Developer — Wirmon

3 months

India

Technologies: PHP, MySQL, HTML, CSS

- Built and delivered end-to-end web applications for multiple clients using PHP and MySQL.
- Developed responsive front ends in HTML and CSS with server-side logic and database design in PHP/MySQL.
- Maintained a 95% client satisfaction rating across delivered projects.

PROJECTS

- **GetFi** (under NDA) — A Telegram mini-app e-commerce platform with gamified rewards, in-app shopping, and an AI-powered conversational recommendation engine (“ChatGPT for shopping”) integrated with the Amazon API. (Telegram Mini App, AI, Amazon API, Full-Stack)
- **Decentralized Land Record Management** — A blockchain-based land management system built for the Government of Odisha to securely record and verify land ownership. (Blockchain, Solidity, Smart Contracts)
- **BlockX Staking App** (under NDA) — A staking decentralized application built on the BlockX chain, with custom staking smart contracts and a React front end. (Solidity, React, BlockX Chain)
- **SuperWorld Map** — A virtual real estate platform where properties are bought, sold, and traded as NFTs on the Polygon network. (MERN, Solidity, Web3.js, Polygon)
- **SuperWorld NFT Salon** — A crypto and NFT marketplace where creators and collectors buy and sell NFTs. (MERN, Solidity, Web3.js)
- **Certichain** — A certificate security and anti-forgery solution using a decentralized IPFS database with on-chain smart contracts. (IPFS, Solidity, Blockchain)
- **DOMX** — A phygital, asset-backed NFT purchase platform enabling investors to buy NFTs via on-ramp payment integration. (Web3, NFT, Payment On-ramp)
- **Distributed Doctor Application** — A distributed health record management system built on blockchain (B.Tech Major Project). (Blockchain, Solidity)
- **Process Lineage** — A blockchain-based process lineage and control tool for supply chain traceability. (Blockchain, Supply Chain)
- **Client Web Application (Wirmon)** (under NDA) — A web application designed and delivered end-to-end for a client, with an HTML/CSS front end and PHP/MySQL back end. (PHP, MySQL, HTML, CSS)

PUBLICATIONS

- COVID-19 Database on Consortium Blockchain — [Google Scholar](#)
- Doctel: Leveraging Blockchain for Secure and Immutable Electronic Health Record Management — [Google Scholar](#)

EDUCATION

B.Tech, Computer Science Engineering — IIIT Bhubaneswar, India

2018 – 2022

CGPA: 8.80 / 10 (Top 3% of batch)

Higher Secondary (XII) — Navy Children School, India

2016 – 2018 • 95%

LEADERSHIP & ACTIVITIES

- **Founder & Advisor**, Metaverse Chapter, IIIT Bhubaneswar
- **Placement Coordinator**, IIIT Bhubaneswar
- **PR Head**, Advaita (College Fest), IIIT Bhubaneswar